MSML602
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Project Progress Report
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Project Title: How Do Weather Conditions Affect WMATA Metrorail Service?

1.Project Overview

Our project takes a first step to examine the effect of weather on WMATA Metrorail service and determines if there are variations in the effect by line, station, and time of day. To answer this question, we plan to create a data set that will incorporate rail incident, weather, scheduled service, and WMATA train information.

The unit of analysis is an event of a train arrival at a particular station, line and direction. The raw WMATA API, however, is designed mainly to give predictions of when a train will arrive, not actual events. So one of the big problems that we have is how to develop a relatable data set from these various sources.

Our pilot was aimed at a number of purposes: to test the process used by us in data collection, to check the quality of the data collected and to look at any problems we should address before we create the final dataset.

2.Pilot Data collection

We initially developed Python scripts to collect data from WMATA APIs and weather sources. Our first run of our collection process only captured a single snapshot of data, making it challenging to assess data quality over time.

To overcome this constraint, we have modified the collection system to enable multiple automated collections. Additionally, we have extended the collection to Train Positions and GTFS-Realtime data.

A pilot was carried out from October 23 to 24, 2026. WMATA Predictions were gathered over a period of about 12 hours and 10 minutes.

Table 1. Pilot Dataset Summary

Data Source	Collected Data
WMATA Predictions	752,791 records
Train Positions	134,108 records
Weather	97 records
Rail Incidents	63 records
Static GTFS	2,176 stops, 6 routes, 14,228 trips, 344,646

	stop times
GTFS-Realtime	3,693 files

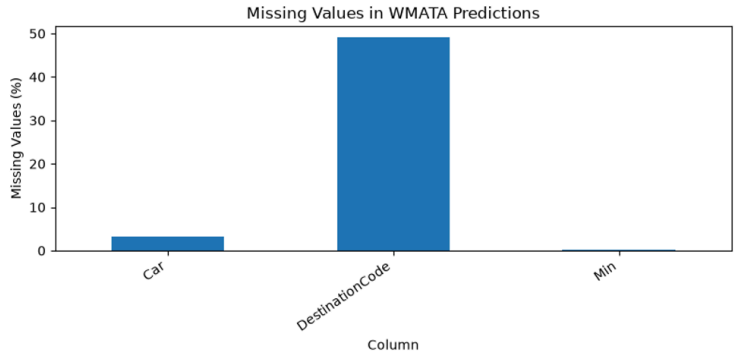
A total of 102 station codes were present in the WMATA Predictions dataset and 1,406 snapshots of collections were available. In the Train Positions, there were 1406 unique snapshots.
We used Python and pandas to explore the structure of the dataset, missing values, duplicated rows, coverage of the data collected, and distribution of key variables for the pilot inspection. Several of our findings we plotted using matplotlib.
No changes were made to the original datasets during the inspection.

3.Initial Data Quality Findings

3.1 Missing and Non-Numeric Values

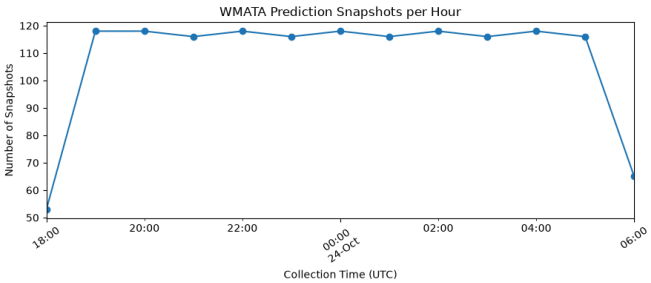
Our inspection identified several missing values in the collected datasets. In WMATA Predictions, DestinationCode had 370,389 missing values(49.20%), while Car had 3.23% missing values. In Train Positions, 5.55% of DestinationStationCode and LineCode values were missing. Some Rail Incidents fields were also completely missing.

The Min Column contained 70,300 non-numberic or missing values, including BRD(45,641), ARR(22,464), and - - -(800). These values will require separated handling during preprocessing. We also found unexpected values such as No and - - in the line.



3.2. Duplicates and Temporal Coverage

We found 252 exact duplicate rows in WMATA Predictions and 114 Train Positions. Weather had 97 records but only 49 unique observation times,while Rail incidents had 63 records but only 4 unique IncidentsIDs. These repeated observations should not be treated as independent events.
Automated collection improved our temporal coverage. WMATA Predictions covered approximately 12 hours and 10 minutes, with 1,406 unique snapshots. The median gap between snapshots was 52.3 seconds, while the largest gap was about 5.1 minutes. This suggests that collection generally worked, although some gaps remained.



3.3 Weather and Other Limitations

No precipitation or rain was recorded during the pilot. Our final dataset will covers 7days, so we need to choose proper 7days that include rainy days to examine the effect of rain on Metro service.

The major Static GTFs files were successfully loaded. We also identified 3,693 GTFS-Realtime files and successfully decompressed a sample file, although the internal Protobuf data was not fully inspected.

Overall, the pilot helped us identify data quality issues that need to be addressed before creating the final dataset.

4.Changes Based on Pilot Findings

There were two major changes as a result of our pilot with regard to our data collection process. First, we improved temporal coverage by automating the process of data collection.

For our first data collection, we were only able to get a single snapshot of data. The new collection system gave us the ability to collect new observations repeatedly, and from the success of the pilot(which was around 12 hours of collection) it appeared that we will be able to collect more observations without having to manually run each collection.

Secondly, we increased the number of data sources.

We added Train Positions and GTF-s Realtime data to our original WMATA prediction, incident, weather and Static GTFS data collection. These other sources might helpful in later stages of identification of train movements and reconstruction of arrival events.

Additionally, the pilot helped to identify some areas where there were not current solutions, such as lack of destination codes, repeated weather observations, and the presence of mixed data in the min column.

In the next step, we will be working to create rules for preprocessing to solve these issues.

5.Data Integration feasibility

We have evaluated the feasibility of integrating WMATA data with GTFS station information and weather data using a 12 hour pilot dataset. The evaluation focused on station code matching and timestamp alignment to determine whether they could be integrated.

5.1 WMATA and GTFS Station Code Matching

Station codes in WMATA Predictions with the station IDs in Static GTFS were compared. We removed the prefix “STN_” from the GTFS station IDs before the comparison. In this experiment, we could find 94 out of 102 station codes matched GTFS station IDs, which is equivalent to 92.16% matching rate.

5.2 WMATA and Weather Timestamp Matching

Additionally, WMATA Predictions and weather observations based on timestamps were compared. Since WMATA Predictions were collected more frequently than weather observations, we used a backward time based merge with a one hour tolerance. We found that all 752791 WMATA Prediction records matched with weather observations, which is equivalent to 100% matching rate. These results suggest that we can combine WMATA Predictions with weather data using timestamps, however the matching rate does not necessarily mean that both datasets were collected at the same time.
We can confidently say that integrating WMATA Predictions with GTFS station information and weather data is feasible. We still need to investigate the unmatched 8 station codes and set up preprocessing rules before we start working on our final dataset.

6.Next Steps

We will be working on a set of rules to handle non-numberic data, repeated observations,and missing values in the preprocessing stage. We will also be developing actual train arrival events and integrating the different data sources.
We are currently using a local scheduler for our collection system. Because it is impractical to keep a local computer running all the time, we will migrate it to a cloud server.
We will use 7 days from the collected data, but not 7 consecutive days, for the final dataset, selecting days with different weather conditions.

NAME	**CONTRIBUTION**
ABDULLAH KHAN	Pilot plan/roadmap, set ups for data and initial tests, data set up and initial formatting/setups
JUNHYUNG HAN	Collected 12hour WMATA pilot data, created a data integration feasibility notebook, and evaluated WMATA GTFS station code matching and WMATA weather timestamp alignment.
DUCKWOO KWON	Developed the initial WMATA data collection code, created a data inspection notebook, and evaluated missing values, duplicates, temporal coverage, and data quality using visualization

AI Usage

DUCKWOO KWON	Codex: Assisted with python syntax ChatGpt: Assisted conducting how to inspect Static GTFS ZIP and GTFS-Realtime Protobuf files, initial data collection.
JUNHYUNG HAN	ChatGPT: assisted with code writing, debugging, and handling trouble shooting.
ABDULLAH KHAN	Helped with comments and summarizing, formatting roadmap, error trouble shooting

INDIVIDUAL REFLECTION
DUCKWOO KWON

1. Tasks Completed :

-Wrote first python scripts to retrieve WMATA Predictions, weather, rail incidents and Static GTFS data.

-I have gathered data from the first collection scripts and taken a snapshot of the data. I used Python and pandas to explore the 12-hour pilot datasets. Examined the characteristics of missing data, duplicate data, temporal coverage, and unexpected data.

- Used matplotlib to create visualizations.

2. Artifacts Produced

- Basic Python data collection scripts: collect_wmata.py, collect_weather.py, collect_incidents.py, collect_gtfs.py and main.py. These were the initial versions, and they have been modified by other team members.

- Data quality and summary statistics, including visualizations, in `WMATA_Data_Inspection.ipynb.

3. Challenges and Decisions

The first collection was only one point in time, so it didn't allow for temporal coverage. In the pilot data inspection, I noticed missing destination codes, duplicate data and non-numeric data for the Min column. The original datasets have not been modified during the inspection and problems were found that needed to be tackled during the preprocessing. I also looked at Static GTFS files and downloaded and unzipped a sample GTFS-Realtime file, but didn't thoroughly look at the internal Protobuf data.

4. Next Steps I will continue to audit quality of data as more data becomes available. During the pilot, I will be working on filling in missing values, removing duplicate records, and removing non-numeric data.

INDIVIDUAL REFLECTION
JUNHYUNG HAN

1. Tasks Completed :

- Executed python script(pilot_12h.py) to collect WMATA Predictions, weather, rail incident data over 12 hour pilot period using APIs
- Uploaded collected datasets(WMATA_12h_pilot.zip) into repository
- Wrote a data integration feasibility analysis(WMATA_Data_Integration_Feasibility.ipynb) using collected datasets
- Updated GitHub repository by uploading project files
- Replaced README content with the 12 hour pilot result

2. Artifacts Produced:

- piilot_12h.py: python script for automated 12 hour data collection
- WMATA_12h_pilot.zip: compressed collected pilot data
- WMATA_Data_Integration_Feasibility.ipynb: Jupyter Notebook for station matching and timestamp integration analysis

3. Challenges and Decisions:

The most challenging part of this project was continuously collecting data for 12 hours. Rather than that, WMATA and Static GTFS station codes were matched 94 out of 102. The unmatched 8 stations are still needed to be preprocessed before integrating the actual datasets. The next data collection will be held in a cloud based environment for convenience and accessibility.

4. Next Steps:

- Preprocess the 8 unmatched WMATA station codes
- Collecting WMATA and weather data over 7 days in a cloud based environment
- Test the feasibility of the collected data
- Upload any updates or suggestions on repository